MATHEMATICS / INDEPENDENT STUDY

MM8036

Type Theory

Progressive hints

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

1 Unit 1 · Binding and substitution

MM8036-U1.01 · E · 15 min
Learning objective: U1.1 Parsing

Fully parenthesize $\lambda x.xyz$ and $ab(cd)$.

Hint 1 — a starting point. Application associates left.

Hint 2 — the key move. The abstraction body extends to the right.

MM8036-U1.02 · E · 15 min
Learning objective: U1.1 Free variables

Compute $\mathrm{FV}(\lambda x.(xy)(\lambda y.yz))$.

Hint 1 — a starting point. Compute each subtree first.

Hint 2 — the key move. The inner binder removes only its own y.

MM8036-U1.03 · E · 15 min
Learning objective: U1.1 Alpha equivalence

Which of $\lambda x.\lambda y.x$, $\lambda a.\lambda b.a$, and $\lambda a.\lambda b.b$ are alpha equivalent?

Hint 1 — a starting point. Track binder positions rather than names.

Hint 2 — the key move. Does the body refer to the outer or inner binder?

MM8036-U1.04 · E · 15 min
Learning objective: U1.2 Substitution

Compute $(\lambda y.xy)[x := yz]$.

Hint 1 — a starting point. The binder conflicts with a free variable of the argument.

Hint 2 — the key move. Rename it to fresh w.

MM8036-U1.05 · C · 30 min
Learning objective: U1.2 Substitution under shadowing

Compute $(\lambda x.(\lambda y.xy)z)[y := x]$ and explain why the replacement has no effect.

Hint 1 — a starting point. Find the free occurrences of y.

Hint 2 — the key move. Every y in this term is bound.

MM8036-U1.06 · C · 30 min
Learning objective: U1.2 Exact free-variable formula

Prove: if $x \in \mathrm{FV}(t)$, then $\mathrm{FV}(t[x := u]) = (\mathrm{FV}(t) \setminus \{x\}) \cup \mathrm{FV}(u)$.

Hint 1 — a starting point. Strengthen the usual inclusion by considering whether the variable occurs.

Hint 2 — the key move. Induct on syntax; in the application case at least one branch contains a free x.

MM8036-U1.07 · C · 30 min
Learning objective: U1.2 Closed substitutions

If t has free variables among $x_1, \ldots, x_n$ and each u_i is closed, prove simultaneous substitution of u_i for x_i closes t.

Hint 1 — a starting point. Use the free-variable bound.

Hint 2 — the key move. Closed replacements cannot introduce new free variables.

MM8036-U1.08 · C · 30 min
Learning objective: U1.2 Composition law

Verify the substitution-composition law for $t = xy$, $u = yz$, $v = w$, substituting first x then y.

Hint 1 — a starting point. Check $x \neq y$ and $x \notin \mathrm{FV}(w)$.

Hint 2 — the key move. Evaluate both sides separately.

MM8036-U1.09 · C · 30 min
Learning objective: U1.1 Structural induction

Let size count one for each variable, application, and abstraction node. Prove $|t[x := u]| = |t| + k(|u| - 1)$, where k is the number of free occurrences of x in t.

Hint 1 — a starting point. Count what happens at each replaced variable leaf.

Hint 2 — the key move. Renaming changes names but not node counts.

MM8036-U1.10 · A · 50 min
Learning objective: U1.2 Necessity of hypotheses

Construct a counterexample to the substitution-composition equation when $x \in \mathrm{FV}(v)$, even though $x \neq y$.

Hint 1 — a starting point. Use $t = y$.

Hint 2 — the key move. Take $u = z$ and $v = x$ with three distinct names.

MM8036-U1.11 · A · 50 min
Learning objective: U1.2 Commuting substitutions

Prove that substitutions for distinct x, y commute if $x \notin \mathrm{FV}(v)$ and $y \notin \mathrm{FV}(u)$: $(t[x := u])[y := v] = (t[y := v])[x := u]$ up to alpha equivalence. Explain why both side conditions are needed.

Hint 1 — a starting point. Apply composition and simplify its replacement term.

Hint 2 — the key move. Use the unchanged-term lemma on $u[y := v]$.

MM8036-U1.12 · A · 50 min
Learning objective: U1.1 Binding invariants

Use zero-based de Bruijn indices to encode the closed terms $\lambda x.\lambda y.x(yx)$ and $\lambda a.\lambda b.a(ba)$. Explain why this representation detects their alpha equivalence, and why a rule for free variables is still needed for open terms.

Hint 1 — a starting point. Index zero denotes the nearest enclosing binder.

Hint 2 — the key move. The body is $1(01)$.

2 Unit 2 · Reduction and confluence

MM8036-U2.01 · E · 15 min
Learning objective: U2.1 Beta reduction

Reduce $(\lambda x.x)(\lambda y.y)$ to normal form.
Hint 1 — a starting point. Locate the outer redex.
Hint 2 — the key move. Substitute the argument for x.

MM8036-U2.02 · E · 15 min
Learning objective: U2.1 Capture avoidance

Reduce $(\lambda x.\lambda y.x)y$.
Hint 1 — a starting point. The free argument must remain free.
Hint 2 — the key move. Rename the inner binder before contraction.

MM8036-U2.03 · E · 15 min
Learning objective: U2.1 Divergence

Let $\Delta = \lambda x.xx$ and $\Omega = \Delta\Delta$. Show an infinite beta sequence from Ω.
Hint 1 — a starting point. Contract its outer redex.
Hint 2 — the key move. The contractum is identical to the source.

MM8036-U2.04 · E · 15 min
Learning objective: U2.1 Reduction strategies

Compare outermost reduction and call-by-value on $(\lambda x.\lambda z.z)\Omega$.
Hint 1 — a starting point. The function ignores its argument.
Hint 2 — the key move. Call-by-value evaluates the argument first.

MM8036-U2.05 · C · 30 min
Learning objective: U2.2 Joining a fork

For $(\lambda x.xx)((\lambda y.y)z)$, contract either the outer redex or the argument redex first and exhibit a common reduct.
Hint 1 — a starting point. The outer step duplicates the argument.
Hint 2 — the key move. Reduce both duplicated argument redexes.

MM8036-U2.06 · C · 30 min
Learning objective: U2.2 Normal-form uniqueness

Derive uniqueness of beta normal forms up to alpha equivalence from confluence.
Hint 1 — a starting point. Apply confluence to two normal reducts of one term.
Hint 2 — the key move. A normal term cannot take a positive number of beta steps.

MM8036-U2.07 · C · 30 min
Learning objective: U2.2 Conversion

Show that beta-convertible normal terms are alpha equivalent. You may use Church–Rosser for conversion.

Hint 1 — a starting point. Conversion can contain reversed steps.

Hint 2 — the key move. Church–Rosser replaces the zigzag by a common reduct.

MM8036-U2.08 · C · 30 min
Learning objective: U2.1 Weak versus strong normalization

Prove that $(\lambda x.\lambda z.z)\Omega$ is weakly normalizing but not strongly normalizing under full beta reduction.

Hint 1 — a starting point. Use two different permitted sequences.

Hint 2 — the key move. Full reduction may reduce inside the ignored argument.

MM8036-U2.09 · C · 30 min
Learning objective: U2.2 Parallel reduction

Compute the complete development of $(\lambda x.xx)((\lambda y.y)z)$. Explain why it differs from contracting just one redex.

Hint 1 — a starting point. Develop the body and argument before substituting.

Hint 2 — the key move. The argument develops to z.

MM8036-U2.10 · A · 50 min
Learning objective: U2.2 Confluence versus termination

Give a finite abstract reduction system that is confluent but has an infinite reduction sequence, and prove both claims.

Hint 1 — a starting point. A self-loop is enough, but add a visible normal form.

Hint 2 — the key move. Use $a \to a$ and $a \to b$ with b normal.

MM8036-U2.11 · A · 50 min
Learning objective: U2.2 Substitution and reduction

Prove that if $u \to u'$, then $t[x := u] \xrightarrow{*} t[x := u']$. Explain why one step is generally insufficient.

Hint 1 — a starting point. Induct on t, renaming binders fresh.

Hint 2 — the key move. There is one copy of the step for each free occurrence of x.

MM8036-U2.12 · A · 50 min
Learning objective: U2.2 Distinguishing convertibility

Prove the closed terms $\lambda x.\lambda y.x$ and $\lambda x.\lambda y.y$ are not beta convertible. Why is their different behavior on some arguments not needed for this proof?

Hint 1 — a starting point. Inspect normal forms and binding structure.

Hint 2 — the key move. Combine normal-form uniqueness for conversion with the de Bruijn invariant.

3 Unit 3 · Simple types and judgments

MM8036-U3.01 · E · 15 min
Learning objective: U3.1 Typing judgments
Give a derivation of $\lambda x.x : A \to A$ in the empty context.
Hint 1 — a starting point. Start with the variable rule.
Hint 2 — the key move. Discharge $x : A$ by abstraction.

MM8036-U3.02 · E · 15 min
Learning objective: U3.1 Application
In context $f : A \to B, x : A$, derive $fx : B$.
Hint 1 — a starting point. Use the variable rule twice.
Hint 2 — the key move. Apply the arrow elimination rule.

MM8036-U3.03 · E · 15 min
Learning objective: U3.1 Arrow association
Give a type and derivation for $\lambda x.\lambda y.x$.
Hint 1 — a starting point. Arrows associate right.
Hint 2 — the key move. First abstract over y, then x.

MM8036-U3.04 · E · 15 min
Learning objective: U3.2 Substitution
Given $x : A \vdash x : A$ and $\vdash u : A$, calculate the conclusion of substitution.
Hint 1 — a starting point. Replace the free variable.
Hint 2 — the key move. The result type remains A in simple typing.

MM8036-U3.05 · C · 30 min
Learning objective: U3.1 Composition
Derive the type of $\lambda f.\lambda g.\lambda x.f(gx)$ from chosen types $f : B \to C, g : A \to B$.
Hint 1 — a starting point. Work from the innermost application outward.
Hint 2 — the key move. Discharge assumptions in reverse order.

MM8036-U3.06 · C · 30 min
Learning objective: U3.2 Weakening
Prove admissibility of adding a fresh unused variable $y : B$ to a simply typed context.
Hint 1 — a starting point. Induct on the typing derivation.
Hint 2 — the key move. Rename abstraction binders to remain fresh for the new context.

MM8036-U3.07 · C · 30 min
Learning objective: U3.2 Preservation

Prove preservation for a root redex $(\lambda x.t)u \to t[x := u]$ using inversion and substitution.
Hint 1 — a starting point. Invert application, then abstraction.
Hint 2 — the key move. The body premise has the exact assumption required by substitution.

MM8036-U3.08 · C · 30 min
Learning objective: U3.2 Open terms and progress

Give a well-typed open term that is neither a call-by-value value nor able to step, under the lecture convention that values are abstractions and the inert base constant.
Hint 1 — a starting point. A neutral application has no visible function constructor.
Hint 2 — the key move. Use $f : o \to o, x : o$.

MM8036-U3.09 · C · 30 min
Learning objective: U3.1 Untypability of self-application

Prove $\lambda x.xx$ is untypable in finite simple types.
Hint 1 — a starting point. Invert the two uses of x in one context.
Hint 2 — the key move. Its type would have to equal an arrow having itself as domain.

MM8036-U3.10 · A · 50 min
Learning objective: U3.2 Failure of subject expansion

Construct a beta step from an untypable term to a typable closed term. Prove both typing assertions.
Hint 1 — a starting point. An ignored argument may itself be untypable.
Hint 2 — the key move. Use $(\lambda x.\lambda y.y)\Omega$.

MM8036-U3.11 · A · 50 min
Learning objective: U3.2 Substitution proof

Prove the application and abstraction cases of the substitution lemma, including why a binder may need renaming.
Hint 1 — a starting point. State the induction hypothesis for each immediate premise.
Hint 2 — the key move. For an abstraction, choose its binder fresh for the substituting term and distinct from the variable being removed.

MM8036-U3.12 · A · 50 min
Learning objective: U3.1 Typing restricts syntax

Prove that a closed beta-normal arrow-only term of base type o, in the language with exactly one inert constant $c : o$, must be c.
Hint 1 — a starting point. Classify normal terms into abstractions and neutral spines.
Hint 2 — the key move. A closed neutral spine must be headed by a constant.

4 Unit 4 · Normalization and constructive logic

MM8036-U4.01 · E · 15 min
Learning objective: U4.2 Curry–Howard
Write a proof term for $A \to B \to A$.
Hint 1 — a starting point. Introduce both assumptions.
Hint 2 — the key move. Return the first assumption.

MM8036-U4.02 · E · 15 min
Learning objective: U4.2 Conjunction
Construct $(A \times B) \to A$ and $(A \times B) \to B$.
Hint 1 — a starting point. Use product elimination.
Hint 2 — the key move. Use the two projections.

MM8036-U4.03 · E · 15 min
Learning objective: U4.2 Disjunction
Construct $A \to A + B$ and explain what data the output contains.
Hint 1 — a starting point. Choose the left constructor.
Hint 2 — the key move. Keep the supplied evidence for A.

MM8036-U4.04 · E · 15 min
Learning objective: U4.1 Normalization vocabulary
State weak and strong normalization and their implication relation.
Hint 1 — a starting point. One asks for a successful path; the other quantifies over all paths.
Hint 2 — the key move. Every infinite path must be excluded for strong normalization.

MM8036-U4.05 · C · 30 min
Learning objective: U4.2 Negation
Construct $(A \to B) \to ((B \to 0) \to (A \to 0))$.
Hint 1 — a starting point. Assume evidence for A and aim at the empty type.
Hint 2 — the key move. Compose the two functions.

MM8036-U4.06 · C · 30 min
Learning objective: U4.2 Sum elimination
Construct $((A \to C) \times (B \to C)) \to ((A + B) \to C)$ and state its two computation cases.
Hint 1 — a starting point. Project the two functions and inspect the sum.
Hint 2 — the key move. Apply the function matching the tag.

MM8036-U4.07 · C · 30 min
Learning objective: U4.1 Candidate closure

Explain why every variable belongs to every reducibility candidate defined in Lecture 7.
Hint 1 — a starting point. Use neutral closure, not a typing assignment.
Hint 2 — the key move. A variable has no immediate reducts.

MM8036-U4.08 · C · 30 min
Learning objective: U4.1 Why strengthen induction

Show that strong normalization of s and t does not imply strong normalization of st.
Hint 1 — a starting point. Use a normal abstraction that duplicates its argument.
Hint 2 — the key move. Take both terms to be $\Delta = \lambda x.xx$.

MM8036-U4.09 · C · 30 min
Learning objective: U4.2 Double negation

Construct $A \to ((A \to 0) \to 0)$ and explain why reversing the arrow requires a different argument.
Hint 1 — a starting point. An assumption of A can be fed to its negation.
Hint 2 — the key move. Return $\lambda n.na$.

MM8036-U4.10 · A · 50 min
Learning objective: U4.1 Canonicity consequence

Using normalization, preservation, and U3.12, prove every closed term of type o reduces to c in the arrow-only calculus with sole constant $c : o$.
Hint 1 — a starting point. Each theorem supplies one step in the argument.
Hint 2 — the key move. Normalize first, preserve the type, then classify the normal form.

MM8036-U4.11 · A · 50 min
Learning objective: U4.2 Constructive equivalence

Construct maps in both directions between $(A + B) \to C$ and $(A \to C) \times (B \to C)$. Show the composites agree pointwise on constructor inputs.
Hint 1 — a starting point. Restrict a function along each injection; reverse by cases.
Hint 2 — the key move. Test the resulting functions at left and right inputs.

MM8036-U4.12 · A · 50 min
Learning objective: U4.1 Normalization proof dependencies

Reconstruct the abstraction case of the fundamental reducibility lemma. Identify exactly where the controlled beta-expansion lemma is used.
Hint 1 — a starting point. The arrow clause needs strong normalization of the abstraction and reducibility of every application.
Hint 2 — the key move. Use two environment extensions: a fresh variable and an arbitrary reducible argument.

5 Unit 5 · Dependent types

MM8036-U5.01 · E · 15 min
Learning objective: U5.1 Pi formation

State the data needed to form $\Pi_{x:A} B(x)$ in context Γ.
Hint 1 — a starting point. The codomain is checked in an extended context.
Hint 2 — the key move. It may mention the new variable x.

MM8036-U5.02 · E · 15 min
Learning objective: U5.1 Dependent application

If $f : \Pi_{x:A} B(x)$ and $a : A$, what is the type of fa?
Hint 1 — a starting point. Substitute the actual argument into the family.
Hint 2 — the key move. The result lies over a.

MM8036-U5.03 · E · 15 min
Learning objective: U5.2 Sigma introduction

What must be supplied to construct a term of $\Sigma_{x:A} B(x)$?
Hint 1 — a starting point. The second component depends on the first.
Hint 2 — the key move. Supply $a : A$ and then $b : B(a)$.

MM8036-U5.04 · E · 15 min
Learning objective: U5.2 Sigma projections

State the types and computation rules for the projections of $p : \Sigma_{x:A} B(x)$.
Hint 1 — a starting point. The second projection's type mentions the first.
Hint 2 — the key move. Evaluate both projections on (a, b).

MM8036-U5.05 · C · 30 min
Learning objective: U5.1 Dependent identity

Derive the type of $\lambda A.\lambda x.x$ with $A : U_0$. Determine the level of its type.
Hint 1 — a starting point. The inner identity is small, but the outer domain is a universe.
Hint 2 — the key move. Use $U_0 : U_1$.

MM8036-U5.06 · C · 30 min
Learning objective: U5.2 Dependent pair map

Given $f : A \to A'$ and $g : \Pi_{x:A}(B(x) \to B'(fx))$, construct a map $\Sigma_{x:A} B(x) \to \Sigma_{y:A'} B'(y)$.
Hint 1 — a starting point. Eliminate the input pair.
Hint 2 — the key move. Send (a, b) to (fa, gab).

MM8036-U5.07 · C · 30 min
Learning objective: U5.1 Dependent composition
Given $f : \Pi_{x:A} B(x)$ and $g : \Pi_{x:A}(B(x) \to C(x))$, derive a term of $\Pi_{x:A} C(x)$.
Hint 1 — a starting point. Work under an arbitrary $x : A$.
Hint 2 — the key move. Apply g at the same index as f.

MM8036-U5.08 · C · 30 min
Learning objective: U5.2 Existential evidence
Explain the computational difference between $\Sigma_{n:\mathrm{Nat}} P(n)$ and $\Pi_{n:\mathrm{Nat}} P(n)$, and give their respective elimination uses.
Hint 1 — a starting point. One stores a witness; one accepts an arbitrary input.
Hint 2 — the key move. Project a Sigma; apply a Pi.

MM8036-U5.09 · C · 30 min
Learning objective: U5.2 Constant families
Show that Sigma and Pi over constant families reproduce ordinary products and arrows.
Hint 1 — a starting point. Write their introduction, elimination, and computation rules.
Hint 2 — the key move. The substitution into a constant family changes nothing.

MM8036-U5.10 · A · 50 min
Learning objective: U5.2 Constructive choice
Construct a function from $\Pi_{x:A} \Sigma_{y:B(x)} R(x, y)$ to $\Sigma_{f:\Pi_{x:A} B(x)} \Pi_{x:A} R(x, fx)$. Explain why this does not introduce classical choice.
Hint 1 — a starting point. Each input already provides a pair at every x.
Hint 2 — the key move. Define f by first projection and its evidence by second projection.

MM8036-U5.11 · A · 50 min
Learning objective: U5.2 Dependent currying
Construct maps between $\Pi_{p:\Sigma_{x:A} B(x)} C(\pi_1(p), \pi_2(p))$ and $\Pi_{x:A} \Pi_{y:B(x)} C(x, y)$. Verify their computation on pair inputs.
Hint 1 — a starting point. Evaluate an uncurried function on a constructed pair.
Hint 2 — the key move. For the reverse, eliminate the input pair.

MM8036-U5.12 · A · 50 min
Learning objective: U5.1 Context dependency
Explain why exchanging $x : A, y : B(x)$ into $y : B(x), x : A$ is generally invalid, although weakening by a fresh declaration is often admissible. Give a concrete family example.
Hint 1 — a starting point. A context must be well formed from left to right.
Hint 2 — the key move. Use $n : \mathrm{Nat}, v : \mathrm{Vec}(D, n)$.

6 Unit 6 · Induction, identity, and universes

MM8036-U6.01 · E · 15 min
Learning objective: U6.1 Addition computation

With addition recursive in its first argument, compute $\mathrm{add}(\mathrm{suc}(\mathrm{zero}), n)$.
Hint 1 — a starting point. Use successor, then zero computation.
Hint 2 — the key move. Keep the variable n untouched.

MM8036-U6.02 · E · 15 min
Learning objective: U6.1 Induction motive

State the motive for proving right-zero addition.
Hint 1 — a starting point. The statement must vary with the induction input.
Hint 2 — the key move. Use an identity type indexed by n.

MM8036-U6.03 · E · 15 min
Learning objective: U6.2 Transport typing

For $p : \mathrm{Id}_{A(a,b)}$ and $u : B(a)$, state the type of $\mathrm{transport}_{B(p,u)}$ and its reflexivity computation.
Hint 1 — a starting point. It moves evidence to the endpoint fiber.
Hint 2 — the key move. At reflexivity the endpoints coincide.

MM8036-U6.04 · E · 15 min
Learning objective: U6.2 Universe levels

If $A : U_1$ and $B(x) : U_0$, find a universe containing $\Pi_{x:A} B(x)$ and one containing $\mathrm{Id}_{A(a,b)}$.
Hint 1 — a starting point. Use maximum for Pi and the base level for identity.
Hint 2 — the key move. Both depend on the level of A.

MM8036-U6.05 · C · 30 min
Learning objective: U6.1 Associativity of addition

Prove addition associative by induction on its first argument.
Hint 1 — a starting point. Write the motive with the other two arguments fixed.
Hint 2 — the key move. Lift the induction equality through successor.

MM8036-U6.06 · C · 30 min
Learning objective: U6.1 Vector append

Describe and type both clauses of append on length-indexed vectors.
Hint 1 — a starting point. Recurse on the first vector, matching the recursion of addition.
Hint 2 — the key move. The nil case length converts from $0 + n$ to n.

MM8036-U6.07 · C · 30 min
Learning objective: U6.2 Symmetry by J

Derive symmetry of identity with an explicit motive and reflexive case.
Hint 1 — a starting point. The desired output reverses endpoints.
Hint 2 — the key move. Set $C(a, b, p) = \mathrm{Id}_{A(b,a)}$.

MM8036-U6.08 · C · 30 min
Learning objective: U6.2 Congruence

For $f : A \to D$, derive $\mathrm{Id}_{A(a,b)} \to \mathrm{Id}_{D(f(a),f(b))}$. Why is this an implication about identity proofs rather than an equality-reflection rule?
Hint 1 — a starting point. Use path induction on the input proof.
Hint 2 — the key move. The reflexive case is reflexivity at $f(a)$.

MM8036-U6.09 · C · 30 min
Learning objective: U6.1 Safe head

Explain why a head function $\mathrm{Vec}(A, \mathrm{suc}(n)) \to A$ needs no nil branch. Can the same total signature be given for all lengths without extra data?
Hint 1 — a starting point. Constructor indices constrain which inputs can inhabit the domain.
Hint 2 — the key move. Nil has length zero.

MM8036-U6.10 · A · 50 min
Learning objective: U6.2 Transport composition

For $p : \mathrm{Id}_{A(a,b)}$ and $q : \mathrm{Id}_{A(b,c)}$, prove transport along their composite agrees propositionally with successive transport on $u : B(a)$.
Hint 1 — a starting point. Induct on the paths, generalizing their endpoints appropriately.
Hint 2 — the key move. After both become reflexivity, both expressions compute to u.

MM8036-U6.11 · A · 50 min
Learning objective: U6.2 Judgmental and propositional equality

Compare the left-zero and right-zero laws for addition. Explain why a proof of right-zero does not make its open expression reduce by definition.
Hint 1 — a starting point. Inspect which argument the recursive definition matches.
Hint 2 — the key move. A variable in the first argument blocks computation.

MM8036-U6.12 · A · 50 min
Learning objective: U6.2 Diagnosing misuse of path induction

A student argues: "Apply J to $p : \mathrm{Id}_{A(a,a)}$, so every such p equals reflexivity." Identify the missing motive issue; explain what J actually supplies.
Hint 1 — a starting point. J varies an endpoint before specializing to loops.
Hint 2 — the key move. The alleged comparison to reflexivity is not well typed when the endpoint varies.

7 Unit 7 · Substitutions and semantics

MM8036-U7.01 · E · 15 min
Learning objective: U7.1 Context substitution
Describe a substitution from Δ to the context $x : A, y : B(x)$.
Hint 1 — a starting point. Supply terms in declaration order.
Hint 2 — the key move. The second term is checked after the first substitution.

MM8036-U7.02 · E · 15 min
Learning objective: U7.1 Identity substitution
Write the identity substitution on $x : A, y : B(x)$.
Hint 1 — a starting point. Use the context's own variables.
Hint 2 — the key move. Check the dependent second component.

MM8036-U7.03 · E · 15 min
Learning objective: U7.1 Composition order
For $\sigma : \Delta \to \Gamma$ and $\tau : \Theta \to \Delta$, state the source and target of $\sigma \circ \tau$ and the substitution equation for a term t over Γ.
Hint 1 — a starting point. Read composition right to left.
Hint 2 — the key move. First reinterpret t over Δ, then over Θ.

MM8036-U7.04 · E · 15 min
Learning objective: U7.2 Meaning explanations
Contrast a canonical natural-number term with a neutral natural-number term.
Hint 1 — a starting point. Canonical forms expose a constructor.
Hint 2 — the key move. A variable need not compute to a visible numeral in an open context.

MM8036-U7.05 · C · 30 min
Learning objective: U7.1 Associativity on components
Prove associativity of context-substitution composition from the term substitution law.
Hint 1 — a starting point. Compare the component corresponding to each declaration.
Hint 2 — the key move. Both composites substitute twice into the same supplied term.

MM8036-U7.06 · C · 30 min
Learning objective: U7.1 Projection and pairing
For context extension $\Gamma \cdot A$, verify $p \circ \langle \sigma, t \rangle = \sigma$ and $q[\langle \sigma, t \rangle] = t$ using tuple substitutions.
Hint 1 — a starting point. Pairing appends a last component.
Hint 2 — the key move. Projection discards that last component; the generic variable selects it.

MM8036-U7.07 · C · 30 min
Learning objective: U7.2 Families of sets

In the set interpretation, describe the meaning of a dependent sum for $A = \{0, 1\}$, $B(0) = \{a\}$, $B(1) = \{b, c\}$. List its elements.

Hint 1 — a starting point. Keep tags identifying the starting fiber.

Hint 2 — the key move. It is a disjoint union of fibers.

MM8036-U7.08 · C · 30 min
Learning objective: U7.2 Dependent products of sets

For the same two fibers as U7.07, list all dependent functions and count them.

Hint 1 — a starting point. Choose one value from each fiber.

Hint 2 — the key move. The value at zero is forced; at one there are two choices.

MM8036-U7.09 · C · 30 min
Learning objective: U7.2 Semantic substitution

Explain why interpreting $t[\sigma]$ agrees with evaluating the interpretation of t in the environment determined by σ. Prove the variable and application cases.

Hint 1 — a starting point. Substitution and evaluation both proceed structurally.

Hint 2 — the key move. For a variable, the tuple component is selected on either side.

MM8036-U7.10 · A · 50 min
Learning objective: U7.2 Soundness is not completeness

The set model interprets every identity proof between fixed endpoints, when one exists, by the same element. Explain why soundness alone does not prove syntactic UIP.

Hint 1 — a starting point. Soundness sends derivable equalities to semantic equalities.

Hint 2 — the key move. Its implication points in only one direction.

MM8036-U7.11 · A · 50 min
Learning objective: U7.1 Explicit substitution design

Why does the equation $(t[\sigma])[\tau] = t[\sigma \circ \tau]$ not by itself prove that an explicit-substitution rewrite system terminates? Construct a simple warning using orientations.

Hint 1 — a starting point. An equality can be oriented in either direction.

Hint 2 — the key move. Allow both directions and a cycle appears.

MM8036-U7.12 · A · 50 min
Learning objective: U7.2 Reindexing a family

Let $A = \{0, 1\}$ with fibers as in U7.07, and let $\sigma : \{r, s, t\} \to A$ send r, s to zero and t to one. Describe the reindexed family, its total space, and its number of sections.

Hint 1 — a starting point. The fiber at an input is the old fiber at its image.

Hint 2 — the key move. Two singleton fibers and one two-element fiber remain.

8 Unit 8 · Categorical models and proof assistants

MM8036-U8.01 · E · 15 min
Learning objective: U8.1 Context morphisms

Explain how contexts and substitutions form a category, naming identities and composition.
Hint 1 — a starting point. Use the equations from Unit 7.
Hint 2 — the key move. Objects are well-formed contexts; arrows supply their variables.

MM8036-U8.02 · E · 15 min
Learning objective: U8.1 Terminal context

Why is the empty context terminal in the substitution category?
Hint 1 — a starting point. Count the declarations that need values.
Hint 2 — the key move. There is exactly one empty tuple from any context.

MM8036-U8.03 · E · 15 min
Learning objective: U8.1 Evaluation

In sets, write evaluation $B^A \times A \to B$ and the curry of $h : X \times A \to B$.
Hint 1 — a starting point. An exponential element is a function.
Hint 2 — the key move. Fix the first coordinate of h.

MM8036-U8.04 · E · 15 min
Learning objective: U8.2 Checker contract

What does successful checking of a lab's theorem establish, and what does it leave to the author?
Hint 1 — a starting point. Separate the encoded type from the informal intention.
Hint 2 — the key move. The checker validates the term against the actual type under its rules.

MM8036-U8.05 · C · 30 min
Learning objective: U8.1 Exponential uniqueness

In sets, prove the curry in U8.03 is the unique $k : X \to B^A$ satisfying $\mathrm{eval} \circ (k \times 1_A) = h$.
Hint 1 — a starting point. Evaluate a candidate at every x, a.
Hint 2 — the key move. The equation forces $k(x)(a) = h(x, a)$.

MM8036-U8.06 · C · 30 min
Learning objective: U8.1 Beta soundness

Explain categorically why interpreting an application of an abstraction satisfies the beta equation.
Hint 1 — a starting point. An abstraction is interpreted by currying.
Hint 2 — the key move. Use the evaluation–curry identity and reindex along the argument.

MM8036-U8.07 · C · 30 min
Learning objective: U8.1 Comprehension in sets

For a set Γ and family $A(\gamma)$, construct the context extension and prove its pairing property.
Hint 1 — a starting point. Use the total space of the family.
Hint 2 — the key move. A point consists of an environment and a value in its fiber.

MM8036-U8.08 · C · 30 min
Learning objective: U8.2 Debugging an induction

An attempted proof of right-zero addition uses reflexivity for the variable case. Explain the failure and repair it using the lab definitions.
Hint 1 — a starting point. The first argument of addition is not a constructor.
Hint 2 — the key move. Split into zero and successor and use congruence.

MM8036-U8.09 · C · 30 min
Learning objective: U8.2 Transport in code

Describe how to cast a vector of length m to one of length n given $p : \mathrm{Id}_{\mathrm{Nat}(m,n)}$. State its reflexive computation.
Hint 1 — a starting point. Use the family of vectors at varying lengths.
Hint 2 — the key move. Transport in $k \mapsto \mathrm{Vec}(A, k)$.

MM8036-U8.10 · A · 50 min
Learning objective: U8.1 CCC and dependent models

Explain why a cartesian closed category by itself supplies a model of simple arrows but does not automatically provide all the dependent-type structure used here.
Hint 1 — a starting point. Simple arrows use one fixed object for each type.
Hint 2 — the key move. Dependent types vary over contexts and must reindex.

MM8036-U8.11 · A · 50 min
Learning objective: U8.2 Constructive proof audit

Construct a program of type $(A + (A \to 0)) \to (((A \to 0) \to 0) \to A)$. Explain both branches, why this does not establish unrestricted double-negation elimination, and why replacing the missing decision input by a postulate would violate the completed-lab contract.
Hint 1 — a starting point. Inspect the supplied decision before using the double negation.
Hint 2 — the key move. In the negative branch, the double negation produces an impossible input to empty-type elimination.

MM8036-U8.12 · A · 50 min
Learning objective: U8.1 Synthesis of substitution and semantics

In the families-of-sets model, prove that reindexing a term along $\sigma : \Delta \to \Gamma$ and then $\tau : \Theta \to \Delta$ equals reindexing along $\sigma \circ \tau$. Relate this to the syntactic explicit-substitution equation.
Hint 1 — a starting point. A term is a section $t(\gamma) \in A(\gamma)$.
Hint 2 — the key move. Evaluate both constructions at an arbitrary θ.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM8036-M1.01 · E · 15 min
Learning objective: U1–U2 Binding and reduction
Compute $\mathrm{FV}((\lambda x.\lambda y.xy)z)$ and reduce the term to normal form.
Hint 1 — a starting point. Compute free variables before contraction.
Hint 2 — the key move. The free z does not conflict with y.

MM8036-M1.02 · E · 15 min
Learning objective: U3–U4 Typing and logic
Derive the type $(A \to B) \to A \to B$ and interpret the term as a logical proof.
Hint 1 — a starting point. Introduce a function and its input.
Hint 2 — the key move. Apply the function to the input.

MM8036-M1.03 · C · 30 min
Learning objective: U2–U4 Normalization
Give a weakly but not strongly normalizing term, then explain why it cannot be typable in the arrow-only core.
Hint 1 — a starting point. Use an ignored divergent argument.
Hint 2 — the key move. Combine the explicit infinite path with strong normalization.

MM8036-M1.04 · C · 30 min
Learning objective: U5–U6 Dependent data
Define vector append on paper and justify its result index in both cases.
Hint 1 — a starting point. Match on the first vector.
Hint 2 — the key move. Its length recursion follows addition on the first argument.

MM8036-M1.05 · A · 50 min
Learning objective: U3–U4 Metatheory synthesis
In the arrow-only language with sole constant $c : o$, show every closed term of type $o \to o$, applied to c, reduces to c. Explain what this does and does not say about equality of functions.
Hint 1 — a starting point. Type the application and use base-type canonicity.
Hint 2 — the key move. Testing one closed argument is not a general extensionality rule.

MM8036-M1.06 · A · 50 min
Learning objective: U5–U8 Proof and model
Construct the dependent-pair map of U5.06 and explain how the same formula acts on total spaces in the set model. Prove it respects the projection to the new base.
Hint 1 — a starting point. The first coordinate determines the target fiber.
Hint 2 — the key move. Project after mapping and compare with applying f after projection.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM8036-M2.01 · E · 15 min
Learning objective: U1–U3 Substitution
Reduce $(\lambda x.\lambda z.xz)y$ and give its type under $y : A \to B$.
Hint 1 — a starting point. No binder conflicts with the argument.
Hint 2 — the key move. The resulting abstraction applies y.

MM8036-M2.02 · E · 15 min
Learning objective: U5–U6 Identity
Give the type and defining reflexive case of transport in a family $P : A \to U_0$.
Hint 1 — a starting point. Specify both endpoints.
Hint 2 — the key move. At reflexivity it is the identity function.

MM8036-M2.03 · C · 30 min
Learning objective: U4–U5 Constructive evidence
Construct a term taking $\Sigma_{x:A}(B(x) \times C(x))$ to $(\Sigma_{x:A}B(x)) \times (\Sigma_{x:A}C(x))$. Explain why the reverse direction is generally unavailable.
Hint 1 — a starting point. Use the same witness in both output pairs.
Hint 2 — the key move. Two independent existential witnesses need not coincide.

MM8036-M2.04 · C · 30 min
Learning objective: U6 Induction
Prove that mapping the identity function over a vector returns the original vector, using propositional equality.
Hint 1 — a starting point. Induct on the vector.
Hint 2 — the key move. Use congruence of the fixed-head cons constructor.

MM8036-M2.05 · A · 50 min
Learning objective: U2–U4 Confluence and typing
Suppose a closed simply typed term reduces along two paths to u, v. Prove u, v have a common normal form of the original type.
Hint 1 — a starting point. Use preservation, normalization, and confluence in different roles.
Hint 2 — the key move. Normalize both reducts and then compare them.

MM8036-M2.06 · A · 50 min
Learning objective: U6–U8 Equality audit
Explain why the following inference is invalid: "The set model identifies all proofs of equality; therefore the paper theory may use K, and the labs may omit `--without-K`." Give a correct relationship among syntax, semantics, and the lab options.
Hint 1 — a starting point. Soundness is one-way, and checker options define the accepted fragment.
Hint 2 — the key move. A model can validate additional principles that the syntax does not assume.